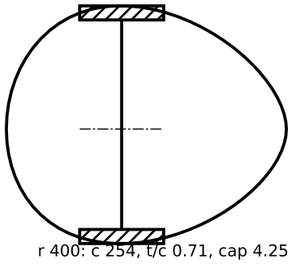
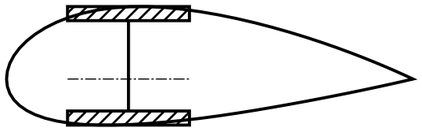
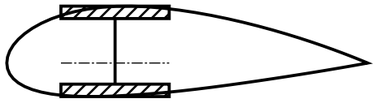
SECTIONS (caps hatched ×3 thickness for clarity, web at 30 % c; LE to the left)



r 400: c 254, t/c 0.71, cap 4.25



r 700: c 368, t/c 0.29, cap 4.25



r 1200: c 327, t/c 0.25, cap 3.75



r 2000: c 260, t/c 0.20, cap 1.93



r 2800: c 194, t/c 0.18, cap 1.00

Zone (r mm)	Shell skin	Core	Spar cap (each side)	Web
280–700 root	2 × 600 g ±45° + 0–12 mm UD/biax build-up	–	merges into root	–
700–1 400	2 × 450 g ±45°	PVC H80 8 mm	UD 1 200 g: 4.25→3.75 mm	2 × 450 g ±45° on 10 mm core
1 400–2 200	2 × 450 g ±45°	PVC H80 6 mm	UD: 3.50→2.25 mm	same
2 200–3 000 tip	2 × 300 g ±45°	PVC H80 5 mm	UD: 2.00→1.00 mm (min)	ends at r 2 900
LE r 2 000–3 000	erosion tape 0.3 mm PU			

NOTES

1. Vacuum infusion, epoxy (Tg ≥ 80 °C, cold-curing then post-cure 8 h at 70 °C), fibre volume ≥ 50 %. Each blade: two shells + web, bonded with structural paste (1–3 mm gaps), leading-edge and trailing-edge flanges 30 mm.
2. Cap thickness follows design6.json t_cap_mm (doc 30 §2); minimum cap 1.0 mm governed by buckling: do not reduce. UD glass E-glass 1 200 g/m² stitched, 0° ± 2° to the pitch axis.
3. Root: laminate builds up from r 700 to a 28 mm wall at the root face around the 10 bonded inserts NS6-105 (fixture NS6-115). Machine the root face square to the pitch axis within 0.2 mm after cure.
4. Gelcoat: polyurethane, white RAL 9010, 0.4 mm. Drain hole Ø4 at the tip trailing edge. Lightning: none on 3 m blades (LPL I, doc 02: tower and nacelle carry it; blade receptor option).
5. Quality: weigh and record mass + static moment; tap test the bonds; one blade per 20 static-tested to SLM H × 1.35 (doc 30).

AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BLADE · SECTIONS AND LAMINATE SCHEDULE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-101	B · 2026-11-06	1:2 / 1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
E-glass/epoxy (infusion), PVC H80 core	20.6 kg	Claude / not checked